Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/05/2015
PI Name:	Douglas, Grace Ph.D.		
Project Title:	Effect of low shear modeled microgravity (LSMMG) on the probiotic Lactobacillus acidophilus ATCC 4356		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	03/21/2014	**End Date:**	09/30/2015
No. of Post Docs:		**No. of PhD Degrees:**	4
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 9/30/2015 per M. Perchonok/JSC HRP (Ed., 6/16/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Castro, Sarah (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Changes in diet, lifestyle, antibiotic usage, and the environmental stresses during spaceflight can alter crewmembers' intestinal microbiome. It has been observed that after two weeks of spaceflight there is a significant decline in the number of bacterial species in the intestine, simultaneous with immune dysregulation and a diminished response to infection. As mission distance and duration increase, the need grows for non-invasive disease prevention and immunomodulation, especially given the limited medical response capability expected for these missions. Based on previous studies that demonstrated enhanced microbiological responses to the low shear environment of spaceflight, the induction of enhanced beneficial responses could logically be expected by probiotic organisms that have demonstrated beneficial interactions with human immune cells. Understanding the microgravity effects on the stress tolerances and genetic expression of probiotic bacteria can enable optimization of growth, survival, strain selection, and conferred benefits in spaceflight. The probiotic Lactobacillus acidophilus has been selected to begin these probiotic studies due to

Task Description:	its wide commercial use and documented benefits that include inhibition of virulence related gene expression in pathogens and mucosal stimulation of immune cells. Ground-based characterization will be accomplished using the low shear modeled microgravity (LSMMG) culture environment in the rotating wall vessel (RWV). Cells cultured within the RWV are maintained in suspension and experience very low levels of fluid shear over their surface, similar to what is expected in microgravity. Numerous investigations have revealed parallel cellular responses to culture within the RWV with liquid culture in spaceflight. Additionally, mathematical modeling has demonstrated that the low levels of fluid shear created by the RWV are physiologically relevant to sites within the human body and provide a more accurate account of the state of probiotic organisms at the brush border of intestinal epithelial cells than would be obtained with standard culture. It is expected that probiotics will be delivered in a rehydratable food product in spaceflight, so liquid culture in an RWV is a logical analog to investigate potential probiotic responses to spaceflight and at human epithelial sites. The objective of this study is to characterize the effect of the low shear modeled microgravity (LSMMG) culture environment in the rotating wall vessel (RWV) on the growth, stress response, and transcriptomic response of L. acidophilus. We hypothesize that L. acidophilus will maintain or improve its growth and stress response following culture in the LSMMG environment, and that transcriptomic analysis will define the associated molecular mechanism(s), resulting in the ability to optimize strain selection. Specific Aim: Determine the effects of LSMMG culture on the growth, stress response, and transcriptomic profile of L. acidophilus ATCC 4356.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal.
Research Impact/Earth Benefits:	
Task Progress:	The introduction of generally recognized as safe (GRAS) probiotic microbes into the spaceflight food system has the potential for use as a safe, non-invasive, daily countermeasure to crew microbiome and immune dysregulation. However, the microgravity effects on the stress tolerances and genetic expression of probiotic bacteria must be determined to confirm translation of strain benefits and to identify potential for optimization of growth, survival, and strain selection for spaceflight. The work completed demonstrates the translation of characteristics of a GRAS probiotic bacteria to a microgravity analog environment. Lactobacillus acidophilus ATCC 4356 was grown in the low shear modeled microgravity (LSMMG) orientation and the control orientation in the rotating wall vessel (RWV) to determine the effect of LSMMG on the growth, survival through stress challenge, and gene expression of the strain. No differences were observed between the LSMMG and control grown L. acidophilus, suggesting that the strain will behave similarly in spaceflight and may be expected to confer Earth-based benefits.
Bibliography Type:	Description: (Last Updated: 11/22/2025)
Articles in Peer-reviewed Journals	Castro-Wallace S, Stahl S, Voorhies A, Lorenzi H, Douglas GL. "Response of Lactobacillus acidophilus ATCC 4356 to low-shear modeled microgravity." Acta Astronaut. 2017 Oct;139:463-8. https://doi.org/10.1016/j.actaastro.2017.07.033 , Oct-2017